Systems working together

Practice Paper 2 | 30 marks | Suggested time: 35 minutes

Name: _____ Class: _____

Instructions: Answer ALL questions. Use full sentences when asked to explain. Show calculations and units. The number in square brackets tells you the marks available.

Question 1 Circulation — 8 marks

a) For arteries, veins and capillaries, state ONE distinctive feature or function of each. **[3]**

b) Describe the route of blood from the right atrium, through the lungs, to the left ventricle. **[3]**

c) Explain why the left ventricle has a thicker muscular wall than the right ventricle. **[2]**

Question 2 Diabetes — 6 marks

a) State where insulin is produced and explain its role in controlling blood glucose. [2]

b) Compare Type 1 and Type 2 diabetes. Include the body's response to insulin and ONE fact about how each condition develops. [4]

Question 3 Interpreting heart-rate data — 6 marks

Students measure heart rate at several points after activity:

Before activity: 68 beats/min; immediately after: 132 beats/min; 2 minutes later: 96 beats/min; 4 minutes later: 74 beats/min.

a) Calculate the increase from before activity to immediately after activity. Give units. [1]

b) Describe how heart rate changes over the four minutes after activity. [2]

c) Explain why heart rate increases during activity using glucose, oxygen, blood and aerobic respiration. **[3]**

Question 4 A blocked coronary artery — 4 marks

a) State the role of the coronary arteries. **[1]**

b) Explain how a coronary artery blockage can cause heart muscle cells to die. **[3]**

Question 5 Extended response: linking the systems — 6 marks

a) A student eats a carbohydrate-rich meal, then walks briskly to school. Explain how the digestive, gas-exchange and circulatory systems supply a leg muscle cell with the substances it needs to release energy. Use a logical sequence and scientific words. **[6]**

End of paper. Check that you have answered all parts. Total: 30 marks.